MODULE 08 · DOMAIN 1 - DATA PREPARATION FOR ML AND AI

Data integrity, bias and compliance

Glue Data Quality, Clarify bias metrics, class imbalance, and protecting sensitive data.

MODULE 08 of 43 DOMAIN 1 - Data Prep WEIGHT 28%

admin@sthithapragna.com

A model trained on flawed data produces **confident, plausible, wrong** answers.

What has to be true before you train

01

Correct

The data says what it claims to say.

02

Fair

It represents the groups the model will serve.

03

Lawful

You are allowed to hold it and use it this way.

04

Recorded

You can show later what you checked and found.

ON THE EXAM

These are separate checks. Data can be perfectly accurate, deeply unrepresentative, and legally problematic all at once.

AWS Glue Data Quality

- Define expectations as **rules**, evaluate them, and get a **data quality score**.
- The score is simply the **percentage of rules that pass**.
- **Recommend rules** analyses your data and proposes a starting ruleset for you.
- A ruleset is **associated with a table in the Glue Data Catalog**.
- Built on Deequ, the open-source data quality library. Generally available since June 2023.

> ON THE EXAM
>
> Named explicitly in the exam guide, and barely present in most question banks. Worth knowing properly.

Six dimensions worth checking

DIMENSION	THE QUESTION, AND A RULE THAT TESTS IT
Completeness	Are required values present? IsComplete
Uniqueness	Are there duplicates where there should not be? IsUnique
Validity	Does the value fit its allowed set or range? ColumnValues in […]
Accuracy	Does it match a trusted reference? Hardest to test automatically
Consistency	Do related fields agree? Order date before delivery date
Timeliness	Is the data recent enough to be worth training on?

ON THE EXAM

Timeliness is the one nobody writes a rule for, and stale training data is a very common cause of a disappointing model.

DQDL: rules and analyzers

CONCEPT	WHAT IT DOES
DQDL	Data Quality Definition Language. Rules live in a list named Rules
Rule	An expectation that returns true or false. IsComplete "passenger_count"
Analyzer	Gathers statistics with no expectation. RowCount, DistinctValuesCount
Dynamic rule	A threshold relative to history. RowCount > avg(last(10))
Why analyzers	For columns you care about but cannot yet write a threshold for

ON THE EXAM

Rules need you to know what good looks like. Analyzers do not, which is why they exist.

ML-powered anomaly detection, and its one limitation

- Learns normal statistical ranges over time and flags **deviations rules would miss**.
- It handles **seasonality**, which static thresholds fundamentally cannot.
- It then **recommends DQDL rules** capturing the pattern it observed.
- **Supported in AWS Glue ETL only. Not in Data Catalog-based data quality.**
- Needs at least **three values per statistic** before it generates an observation.

> ON THE EXAM
>
> Generally available since August 2024, so most question banks predate it entirely.

What happens when a rule fails

- **Fail the job.** Nothing downstream runs on data you do not trust.
- **Quarantine the bad rows** and continue with the rest. Usually the pragmatic choice.
- **Continue and record** the score, when the check is informational.
- Results can raise **EventBridge** events, so alerting and pipeline stops are automatic.
- A quality score nobody acts on is documentation of a problem, not a control.

> ON THE EXAM
>
> Deciding the failure action is the real design work. Writing the rule is the easy half.

Bias is measurable **before a model exists**.
That is what surprises people.

Facets, and what bias means here

- A **facet** is the column bias is measured against. Gender, age band, region.
- **Facet a** is the advantaged group. **Facet d** is the disadvantaged group.
- Bias here is a **statistical property of the data**, not an accusation about anyone.
- Clarify measures and reports. It does not fix anything automatically.
- You also specify which label value counts as the positive outcome.

> ON THE EXAM
>
> You cannot fix what you cannot see. Clarify's job is to make the disparity visible and numeric.

SageMaker Clarify is in maintenance mode

- **Closed to new customers from 30 July 2026.** Existing customers continue unaffected.
- It remains named in the exam guide, in Tasks 1.3, 2.3 and 4.1. **Still examinable.**
- The **metrics themselves are not proprietary**. CI, DPL, KL, JS and the rest are published.
- AWS points to open-source monitoring solutions and **SageMaker AI MLflow Apps** as replacements.
- For foundation models, **Amazon Bedrock Evaluations** is the named path.

> ON THE EXAM
>
> Learn the metrics, not just the service. The mathematics outlives the console that computed it.

Pre-training bias metrics

METRIC	THE QUESTION IT ANSWERS
CI Class Imbalance	Is one group under-represented in the data at all?
DPL Diff. in Positive Proportions in Labels	Do positive outcomes differ between the groups?
KL Kullback-Leibler Divergence	How far apart are the label distributions? Asymmetric
JS Jensen-Shannon Divergence	The same, symmetric. Order of the groups does not matter
LP L-p Norm	Norm between the label distributions. L2 by default
TVD Total Variation Distance	Half the L1 distance between the distributions
KS Kolmogorov-Smirnov	The maximum difference across label values
CDDL Conditional Demographic Disparity	Disparity after controlling for another variable

ON THE EXAM

Five of these are variations on 'how different are the two label distributions'. Two are not, and those two matter most.

What a Clarify bias job needs

CONFIGURATION	WHAT YOU SUPPLY
DataConfig	Input S3 path, output S3 path, the label column, dataset type
BiasConfig	The facet column, and which label values count as positive
Methods	Which metrics to compute: CI, DPL, KL, JS, LP, TVD, KS
Compute	A processing job. Instance type and count, like any Processing job
Output	analysis.json in S3, plus a human-readable report

ON THE EXAM

It runs as a Processing job, so everything from module 04 about Processing jobs applies here too.

Class Imbalance, and its range

- Measures whether facet d has **fewer training samples** than facet a.
- **Range is [−1, 1], not [0, 1].** Plenty of secondary sources get this wrong.
- **Positive** means facet a has more samples. **1** means the data contains only facet a.
- Values near either extreme carry substantial risk of biased predictions.
- Models fit the larger group preferentially, so the smaller group gets higher error.

> ON THE EXAM
>
> The mechanism is worth understanding: a model minimises average error, and average error is dominated by the majority group.

Class Imbalance or DPL?

CI ANSWERS

Are they represented?

- Counts rows in each group.
- Ignores labels entirely.
- Fixed by collecting or resampling data.
- Can be zero while outcomes are unfair.

DPL ANSWERS

Are outcomes equal?

- Compares positive-outcome rates.
- Labels are the whole point.
- Fixed by examining the labelling process.
- Can be large with perfect representation.

THE TIEBREAKER

Equal numbers of applicants, unequal approval rates: CI near zero, DPL large. That combination is a favourite question shape.

KL, JS, LP, TVD and KS in one slide

- All five compare the **label distribution** of facet a with that of facet d.
- **KL is asymmetric.** Swapping the groups changes the answer.
- **JS is the symmetric form of KL.** That is its whole reason for existing.
- **TVD is half the L1 distance. KS is the maximum** difference across label values.
- LP defaults to the L2, or Euclidean, norm between the distributions.

> ON THE EXAM
>
> If a question needs a symmetric measure, that is JS. If it needs the single worst gap, that is KS.

Conditional Demographic Disparity, and Simpson's paradox

- Measures disparity **after controlling for another variable**, such as department or product.
- It exists because aggregate figures can **reverse** when you look within subgroups.
- A dataset can look biased overall and be fair in every subgroup, or the reverse.
- You supply the group variable to condition on. Choosing it is a judgement call.
- The classic illustration is university admissions rates by department.

> ON THE EXAM
>
> This is the metric that stops you drawing a confident conclusion from an aggregate number.

What to actually do about label bias

- **Investigate the labelling process first.** DPL usually reflects historical decisions, not noise.
- **Reweight** examples so the under-favoured group carries more influence during training.
- **Collect more data** for the affected group, if representation is also weak.
- **Dropping the facet column does not work.** Models find proxies: postcode, name, job title.
- Sometimes the right answer is that the dataset is not fit for this purpose.

> ON THE EXAM
>
> Removing gender and leaving postcode in is the classic failed fix. The information is still there, just harder to audit.

Frameworks, and what each one forces

FRAMEWORK	THE OBLIGATION THAT SHAPES YOUR ARCHITECTURE
GDPR	Lawful basis, purpose limitation, erasure rights, data minimisation
HIPAA	Protected health information. BAA required, strict access control and audit
PCI DSS	Cardholder data. Encryption, segmentation, no storage of some fields at all
Sector rules	Financial services often add explainability and retention requirements
Common thread	Know what you hold, restrict who reaches it, and prove both later

ON THE EXAM

You are not examined on legal texts. You are examined on the architecture the obligations force.

Purpose limitation, and why training is a new purpose

- Data collected to deliver a service was not necessarily collected to train models on.
- **Training is a distinct purpose.** Consent for one is not consent for the other.
- A model is also difficult to un-train. Erasure of a row does not erase its influence.
- Practical answer: **record the lawful basis for training** alongside the dataset version.
- This is why module 06's dataset versioning and module 27's model cards exist.

> ON THE EXAM
>
> 'We already have the data' is an access statement, not a permission statement. The exam rewards knowing the difference.

Pre-training bias, and post-training bias

- **Pre-training** metrics look at data and labels. No model required.
- **Post-training** metrics compare the model's **predictions** across facets.
- **FT, the counterfactual fliptest**, checks whether similar examples across groups get similar predictions.
- A dataset with no measurable pre-training bias can still yield a biased model.
- Post-training bias belongs to Domain 2. Module 19 covers it.

> ON THE EXAM
>
> Clarify runs at both stages. A question naming predictions rather than labels is asking about the post-training half.

Fraud is 0.3 percent of transactions. A model that predicts 'not fraud' is 99.7 percent accurate.

Techniques, and what each costs

TECHNIQUE	TRADE-OFF
Random oversampling	Duplicates minority rows. Simple, and encourages overfitting
Random undersampling	Drops majority rows. Fast, and throws away real information
SMOTE	Synthesises new minority points between neighbours. Better than duplicating
Class weights	Penalises minority errors more. No data changes at all
Threshold adjustment	Move the decision boundary after training. Nothing is resampled
Collect more data	The honest answer, when it is possible

ON THE EXAM

Class weights and threshold adjustment change the model's behaviour without touching the dataset. Often the better first move.

Resample the training split only

- **Never resample before splitting.** Synthetic points leak across the split.
- SMOTE before splitting puts interpolated neighbours of test rows into training.
- **Validation and test must keep the real distribution**, or your metrics are fiction.
- A balanced test set tells you nothing about performance on real traffic.
- Same discipline as module 05: split first, then transform.

> ON THE EXAM
>
> Resampling the test set is how a fraud model reports 95 percent recall in testing and catches almost nothing live.

You have to be allowed to hold this data, and to use it this way.

Amazon Macie

- Discovers and classifies **sensitive data in Amazon S3**, using managed data identifiers.
- Detects names, addresses, card numbers, credentials, health identifiers and more.
- **Custom data identifiers** use your own regex, for internal formats it cannot know about.
- Findings go to **EventBridge and Security Hub**, so remediation can be automatic.
- It scans S3. It does not scan a database, or a training job's memory.

> ON THE EXAM
>
> The common surprise is personal data sitting in a data lake nobody classified. Macie is how you find that before a model learns it.

Fine-grained access with Lake Formation

- IAM and bucket policies control access at the **object or prefix** level.
- **Lake Formation** adds table, column, row and cell-level permissions on catalogued data.
- So a data scientist can query a table while **never seeing the salary column**.
- Permissions are granted centrally and enforced across Athena, Glue and Redshift Spectrum.
- Named in the in-scope services list, and easy to forget in an ML context.

> ON THE EXAM
>
> Column-level permissions are often a better answer than making a second, redacted copy of the dataset.

The audit trail you will be asked for

SERVICE	WHAT IT RECORDS
AWS CloudTrail	Who called which API, when, from where. The who and when
AWS Config	How resources were configured over time, and whether that drifted
Glue Data Quality results	What the data looked like on the day you trained
Clarify bias reports	What disparity existed, measured before training
Model cards and lineage	Which dataset version produced which model. Module 27

ON THE EXAM

An audit asks what you knew and when. Reports you generated and kept are the only credible answer.

Techniques, and what survives each one

TECHNIQUE	WHAT YOU KEEP, WHAT YOU LOSE
Redaction	Value removed entirely. Nothing survives, nothing can be recovered
Masking	Partial value kept, such as last four digits. Format survives
Hashing	Same input gives same output. Joins work, the value does not
Tokenisation	Replaced with a token, reversible through a secure lookup
Generalisation	Precision reduced. Exact age becomes an age band
Synthetic data	Statistically similar records, no real individual behind them

ON THE EXAM

Hashing is the one people over-trust: a hashed postcode is still unique, so it can still identify someone in combination.

Synthetic data, honestly assessed

- Generates statistically similar records with **no real individual behind them**.
- Useful for development environments, demos, and sharing with third parties.
- Also fills gaps: SMOTE is a narrow form of synthetic minority data.
- It **inherits the biases** of the data it learned from. It does not launder them.
- It cannot support conclusions about real people, which rules it out for clinical research.

> ON THE EXAM
>
> Synthetic data solves the privacy problem and none of the representativeness problems. Treat it as a tool, not an escape.

Anonymisation or pseudonymisation?

- **Anonymised** data cannot be linked back to a person by anyone, ever.
- **Pseudonymised** data can, if you hold the key or the lookup table.
- Tokenisation and hashing produce pseudonymised data, not anonymised data.
- Under GDPR, pseudonymised data **is still personal data** and keeps its obligations.
- Combining several 'anonymous' columns can re-identify individuals. Watch quasi-identifiers.

> ON THE EXAM
>
> Teams routinely describe pseudonymised data as anonymous. That mistake changes what the law requires of them.

At rest, in transit, and who holds the key

- **SSE-S3** encrypts with AWS-managed keys. On by default, and invisible.
- **SSE-KMS** uses a KMS key, giving you rotation, policies and a CloudTrail audit trail.
- A **customer-managed key** is what compliance requirements usually mean by 'our own key'.
- In transit is TLS. Training jobs can also encrypt **inter-container traffic**.
- Volumes, model artefacts and notebooks all need encrypting too, not just the input bucket.

> ON THE EXAM
>
> The audit trail is often the real requirement. SSE-S3 protects the data; SSE-KMS proves who accessed the key.

Residency, and the constraints it creates

- Some data may not leave a country or region. That is a hard architectural constraint.
- Recall module 02: the training job must be in the **same Region** as its data.
- So a residency rule decides where training runs, not just where storage sits.
- **SCPs** in AWS Organizations can deny actions outside approved Regions outright.
- Check Region availability early. Not every service exists in every Region.

> ON THE EXAM
>
> Residency propagates: storage, training, endpoints, logs and backups all have to obey it, and logs are the one forgotten.

What these controls cost

CONTROL	COST SHAPE
Glue Data Quality	Per DPU-hour of the evaluation. Anomaly detection adds roughly one DPU per statistic
Clarify bias job	Per instance-hour of the Processing job it runs on
Amazon Macie	Per GB of data inspected, plus per S3 object monitored
KMS	Per key per month, plus per API request. High-volume decryption adds up
Lake Formation	No charge for permissions. You pay the underlying query engines

ON THE EXAM

Anomaly detection on a table with thousands of statistics is the surprise here. Choose analyzers deliberately.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
Model accurate overall, poor for one group	Class imbalance on the facet. Check CI
Groups equally represented, outcomes still differ	Label bias. Check DPL, not CI
Bias looks severe overall, fine per department	Simpson's paradox. Check CDDL
Fraud recall excellent in test, poor in production	Resampling applied before the split
Anomaly detection unavailable on a catalogue table	It is supported in Glue ETL only
PII discovered in a training set after deployment	No Macie scan before ingestion

ON THE EXAM

Rows two and three are the ones that separate candidates. Both are cases where the obvious metric gives the wrong story.

Check the data **before** the model exists. Afterwards, every problem looks like a model problem.

A team registers their curated tables in the Glue Data Catalog and configures Data Quality rulesets against them. The rules work.

They now want to catch gradual drifts that no fixed rule anticipates, such as a slow rise in null rates. They enable anomaly detection on the catalogue-based data quality configuration, and it does not become available.

Why, and what should they do?

A Anomaly detection requires at least ten observations, so it will appear after ten runs

B Anomaly detection requires dynamic rules, which must be defined before enabling it

C Anomaly detection is supported in AWS Glue ETL only. Move the evaluation into an ETL job

D Anomaly detection requires the table to use Apache Iceberg format

◆ PAUSE AND DECIDE

Why, and what should they do?

A Anomaly detection requires at least ten observations, so it will appear after ten runs

Mixes in a real number wrongly. Observations need a minimum of three values per statistic, and that governs findings, not availability.

B Anomaly detection requires dynamic rules, which must be defined before enabling it

Dynamic rules and anomaly detection are separate features. Neither is a prerequisite for the other.

C Anomaly detection is supported in AWS Glue ETL only. Move the evaluation into an ETL job

Documented limitation. Configure rules and analyzers in a Glue ETL job and enable anomaly detection there.

D Anomaly detection requires the table to use Apache Iceberg format

Borrows a real concept from module 06 and puts it somewhere it does not apply. Table format is not the constraint.

THE SIGNAL

Two Glue Data Quality entry points exist, and they do not have identical capabilities. That asymmetry is the question.

A bank audits a loan dataset before training. The applicant pool is almost exactly half men and half women, and the team confirms both groups have nearly identical row counts.

However, historical approval rates in the labels are 71 percent for one group and 48 percent for the other. The team must quantify this for the regulator.

Which pre-training bias metric captures this?

A Difference in Positive Proportions in Labels (DPL)

B Class Imbalance (CI)

C Counterfactual Fliptest (FT)

D Conditional Demographic Disparity in Labels (CDDL)

◆ PAUSE AND DECIDE

Which pre-training bias metric captures this?

A Difference in Positive Proportions in Labels (DPL)

Exactly this. It compares the rate of positive outcomes between facets, which is the 71 against 48 percent gap the regulator cares about.

B Class Imbalance (CI)

Counts rows and ignores labels. With near-identical row counts it would report close to zero, missing the disparity entirely.

C Counterfactual Fliptest (FT)

A post-training metric. It compares model predictions, and no model has been trained yet.

D Conditional Demographic Disparity in Labels (CDDL)

The right family, and it answers a further question: does the gap survive controlling for income or loan size? Useful next, not first.

THE SIGNAL

The equal row counts were planted to make CI tempting. Representation is fine here; outcomes are not.

A healthcare analytics team wants to share a training dataset with an external research partner. The data sits in S3 and was assembled from several source systems over three years.

Nobody is certain which columns contain patient identifiers, and free-text clinical notes are included. The partner must be able to join records across files for the same patient.

What should the team do first, and then?

A Redact every column flagged as sensitive, then share the result

B Hash every column, then share, since hashed data is anonymised

C Generate a fully synthetic dataset and share that instead of the real data

D Run Macie to discover sensitive data, then tokenise identifiers and redact free text

◆ PAUSE AND DECIDE

What should the team do first, and then?

A Redact every column flagged as sensitive, then share the result

Removes the identifiers and the ability to join, which the partner explicitly needs. It also skips discovery, so 'flagged' assumes work nobody has done.

B Hash every column, then share, since hashed data is anonymised

The premise is wrong. Hashing is pseudonymisation, not anonymisation, and hashing free-text notes destroys the research value entirely.

C Generate a fully synthetic dataset and share that instead of the real data

Strong for privacy and weak for clinical research, where conclusions must rest on real patients. Also skips discovery.

D Run Macie to discover sensitive data, then tokenise identifiers and redact free text

Discovery first, because nobody knows what is there. Tokenisation preserves joins; free text cannot be reliably tokenised, so it goes.

THE SIGNAL

Two requirements pull against each other: protect identifiers, but preserve joins. Tokenisation is the technique that satisfies both.

A payments team is building a fraud classifier. Fraud is 0.4 percent of the 40 million labelled transactions. An engineer proposes applying SMOTE to the full dataset until the classes are balanced, then splitting 80/20 for training and testing.

They expect this to raise recall substantially and plan to report accuracy as the headline metric.

What is wrong with this plan?

A SMOTE is unsuitable for tabular data and should be replaced with random oversampling

B Resampling before the split leaks synthetic points across it, and accuracy is the wrong metric

C 40 million rows is too many for SMOTE, so the majority class should be undersampled first

D Nothing is wrong. Balancing before splitting gives both sets the same distribution

◆ PAUSE AND DECIDE

What is wrong with this plan?

A. SMOTE is unsuitable for tabular data and should be replaced with random oversampling

Backwards. SMOTE was designed for tabular data, and random oversampling duplicates rows, which overfits more rather than less.

B. Resampling before the split leaks synthetic points across it, and accuracy is the wrong metric

Two faults, both real. Synthetic minority points interpolate from rows that land in test, and 99.6 percent accuracy is achievable by predicting 'not fraud'.

C. 40 million rows is too many for SMOTE, so the majority class should be undersampled first

Raises a genuine scale concern and ignores both actual faults. Undersampling first would still leave the leak and the metric intact.

D. Nothing is wrong. Balancing before splitting gives both sets the same distribution

It does, and that is precisely the problem. A balanced test set measures performance on a world that does not exist.

THE SIGNAL

Note D's reasoning: it states the outcome correctly and treats it as the benefit. A balanced test set is the fault, not the feature.

What to take from this module

- **CI asks about representation. DPL asks about outcomes.** They are not interchangeable.
- **JS is symmetric KL. KS is the maximum gap. CDDL controls for another variable.**
- **Resample the training split only.** A balanced test set produces fictional metrics.
- **Glue anomaly detection is ETL only**, not Data Catalog-based data quality.
- **Hashing is pseudonymisation, not anonymisation.** The obligations stay with you.

> ON THE EXAM
>
> Module 09 begins the genuinely new C02 material: embeddings and vector stores.